NOTE

All numerical values are in metric units [with U.S. customary units in brackets]. Dimensions are in millimeters. Unless otherwise specified, dimensions have a tolerance of ±0.13 and angles have a tolerance of ±2°. Figures and illustrations are for identification only and are not drawn to scale.

DANGER

HIGH-VOLTAGE - MORTAL DANGER

- *This connector is intended for use in high-voltage applications. Special care must be applied to ensure that the connector functions as intended.*
- *If you suspect that the connector has been modified, damaged, contaminated, or otherwise compromised, please discontinue its use immediately.*
- *This connector should only be serviced by a trained and qualified technician.*

1. INTRODUCTION

This specification covers the requirements for application of the Sealed Two-Position AMP+ HVA280-2phm Plug High-Voltage Connector. The AMP+ HVA280 system is designed for a metric wire size range of 2.5 mm² to 4 mm². The connector incorporates conductive EMI shields to reduce radiated emissions in the application.

The AMP+ HVA280 connector system is available in five different keying or polarizing configurations with an integrated High Voltage Inter-Lock (HVIL) mechanism. Un-mating of the connector system is a multiple step process which enables the system to meet a delayed un-mate sequence.

The AMP+ HVA280 connector system incorporates up to two 2.8 mm receptacle contacts and a 0.63 mm shunt in the plug connector which mate to up to two 2.8 mm blade contacts and two 0.63 mm receptacle contacts in the header assembly. Termination tooling is available for both types of contacts. The connector housing is molded in orange to denote a high voltage system.

CAUTION

These high-voltage connectors must NOT be mated with any other type of connector.

When corresponding with TE Connectivity Personnel, use the terminology provided in this specification to facilitate your inquiries for information. Basic terms and features of this product are provided in Figure 1.

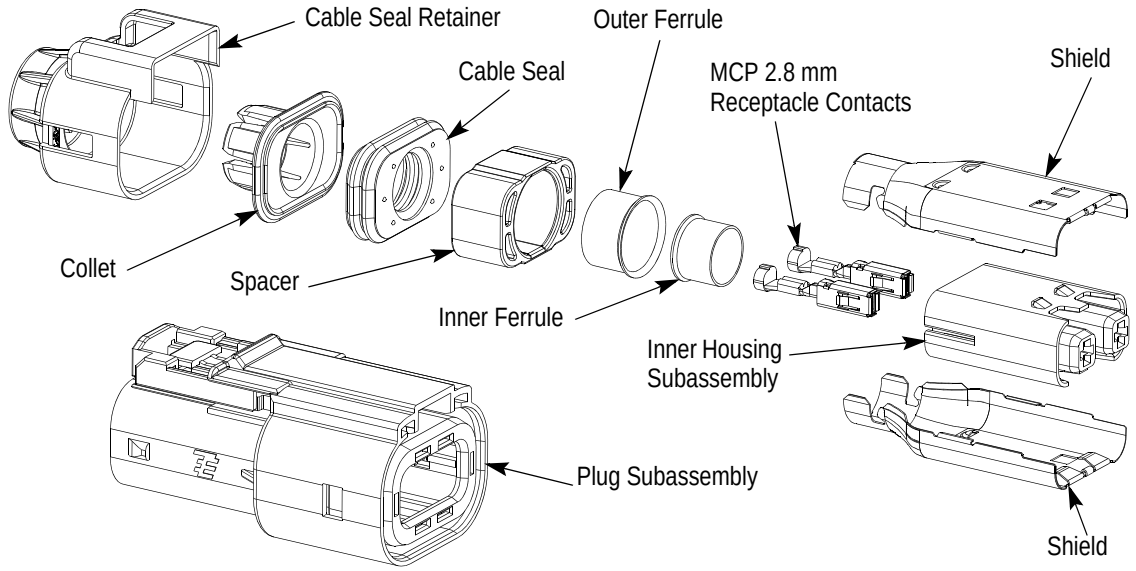


Figure 1

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2. REFERENCE MATERIAL

2.1. Revision Summary

- Updated document to corporate requirements
- Added and or deleted text in Paragraphs 2.5, 3.6.1, and 3.6.9
- Added new Figure 10 and renumbered
- Added new art and related text to new Figure 24

2.2. Customer Assistance

Reference Base Product Part Numbers 2103014, 2103437 and Product Code K994 are representative numbers of the Sealed Two-Position AMP+ HVA280-2phm Plug High-Voltage Connector. Use of these numbers will identify the product line and expedite your inquiries through a service network established to help you obtain product and tooling information. Such information can be obtained through a local TE Representative or, after purchase, by calling the Product Information Center at the number at the bottom of page 1.

2.3. Drawings

A. Customer Drawings

Customer Drawings for product part numbers are available from the service network. If there is a conflict between the information contained in the Customer Drawings and this specification or with any other technical documentation supplied, the Customer Drawing takes preference.

B. Reference Drawings

NOTE

Reference Drawings (C-2103014) and (C-2103437) are available for part number relationships only and are not saleable items. Refer to these drawings only for compatible part numbers for customer inquiry. For more information on Reference Drawings, call PRODUCT INFORMATION at the number at the bottom of page 1.

2.4. Specifications

Application Specification 114-18148-1 provides termination procedures and application requirements for the 1-968851-3 and 1-968853-3 MCP 2.8 mm receptacle terminals. Application Specification 114-18387 provides termination and application requirements for the 1241390-3 MCP 2.8K receptacle terminals.

2.5. Cable and Subcomponent Specification (Figure 2)

The following cable and subcomponent combinations have been validated by TE and should be used together to ensure optimum connector performance. Alternative cables may be used with the AMP+ HVA280 connector after ensuring performance through validation testing.

Cable Option

Cable Supplier: COROPLAST
Cable Construction: 2X 4.0 mm^2
Supplier Cable Part Number: 9-2641

Recommended Subcomponents:

1-968853-3 MCP Receptacle Contact, 2.8 mm
2103157-1 or 1-2103157-1 Shield, Plug
1587723-3 Inner Ferrule
1587724-3 Outer Ferrule
2103154-1 Cable Seal
2103013-1 Cable Seal Retainer
2103155-1 Collet

Cable Option

Cable Supplier: COROPLAST
Cable Construction: 2X 2.5 mm^2
Supplier Cable Part Number: 9-2641

Recommended Subcomponents:

1241390-3 or 1-968851-3 MCP Receptacle Contact, 2.8 mm
1-2103157-2 Shield, Plug
1587723-2 Inner Ferrule
1587724-2 Outer Ferrule
2103154-3 Cable Seal
2103013-2 Cable Seal Retainer
2103155-3 Collet

COROPLAST is a trademark.

Cable Option

Cable Supplier: DRAKA
Cable Construction: 2X 4.0 mm^2
Supplier Cable Part Number: FLR91XBC33X

Recommended Subcomponents:

1-968853-3 MCP Receptacle Contact, 2.8 mm
1-2103157-2 Shield, Plug
1587723-2 Inner Ferrule
1587724-2 Outer Ferrule
2103154-3 Cable Seal
2103013-2 Cable Seal Retainer
2103155-3 Collet

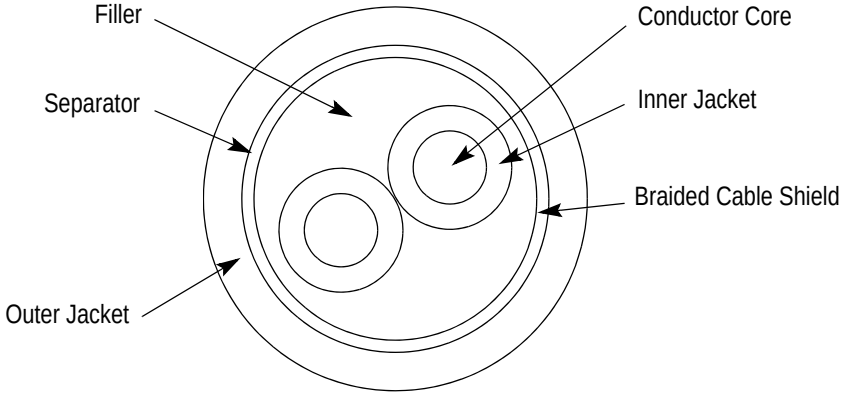


Figure 2

2.6. Instructional Material

The following list includes available instruction sheets (408-series) that may provide assembly procedures for product, operation, maintenance and repair of tooling, and customer manuals (409-series) that provides setup, operation, and maintenance of machines.

Document Number	Document Title
408-2498	Crimping Head Cross Reference for Pneumatic Tools
408-4070	Pneumatic Head Assembly 679304-1
408-4303	Pneumatic CERTI-CRIMP* Tool Holder Assembly 356302-1
408-7424	Checking Terminal Crimp Height Gaging Die Closure
408-9930	PRO-CRIMPER* III Hand Crimping Tool Frame Assembly 354940-1
408-10441	Sealed Two-Position HVA280 Header High-Voltage Connector System
409-5862	626 Pneumatic Tooling Assemblies 189721-[ ] and 189722-[ ]
411-18087	ERGOCRIMP* Hand Tool 539635-1
411-18169	ERGOCRIMP Die 539723-2

3. REQUIREMENTS

3.1. Safety

Do not stack contact packages so high that the shipping containers buckle or deform.

3.2. Shelf Life

The contacts should remain in the shipping containers until ready for use to prevent deformation to the contacts and/or damage to the housings. The products should be used on a first in, first out basis to avoid storage contamination that could adversely affect signal transmissions.

3.3. Chemical Exposure

Do not store contacts near any chemicals listed below, as they may cause stress corrosion cracking in the contacts.

Alkalies	Ammonia	Citrates	Phosphates Citrates	Sulfur Compounds
Amines	Carbonates	Nitrites	Sulfur Nitrites	Tartrates

DRAKA is a trademark.

3.4. Material

The connector assemblies are made of thermoplastic materials and the terminals and shields are tin and silver-plated copper alloy.

3.5. Circuit Identification

The terminal cavities are numbered on the inner housing of each connector at the wire entry end.

3.6. Assembly Procedures

The following procedures show the details of the cable assembly and insertion instructions of the cable assembly into the plug subassembly.

NOTE

Refer to Figure 2 for the various components of the cable as shown in an end view of the cable.

1. In the orientation shown in Figure 3, slide the cable seal retainer onto the cable first, then the collet, cable seal, spacer, then finally the outer ferrule.

NOTE

Optional: The appropriate collet tool listed in Figure 24 can be used to ease sliding the collet down the cable.

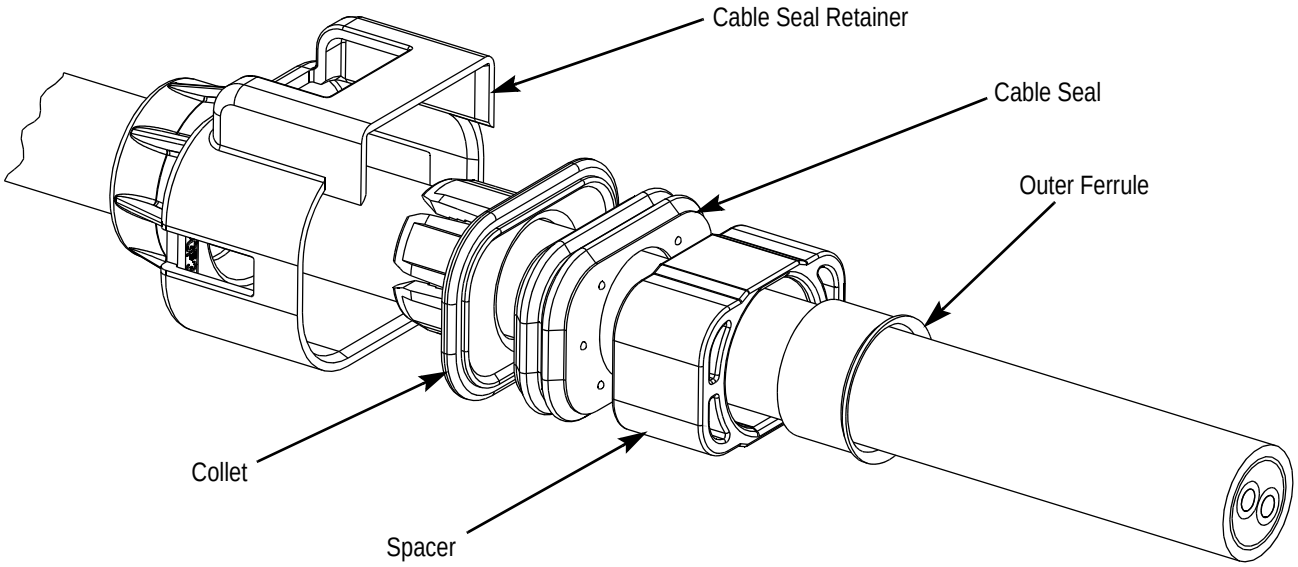


Figure 3

2. Strip and remove the outer cable jacket and separator (if present) from the end, exposing the braided cable shield as shown in Figure 4.

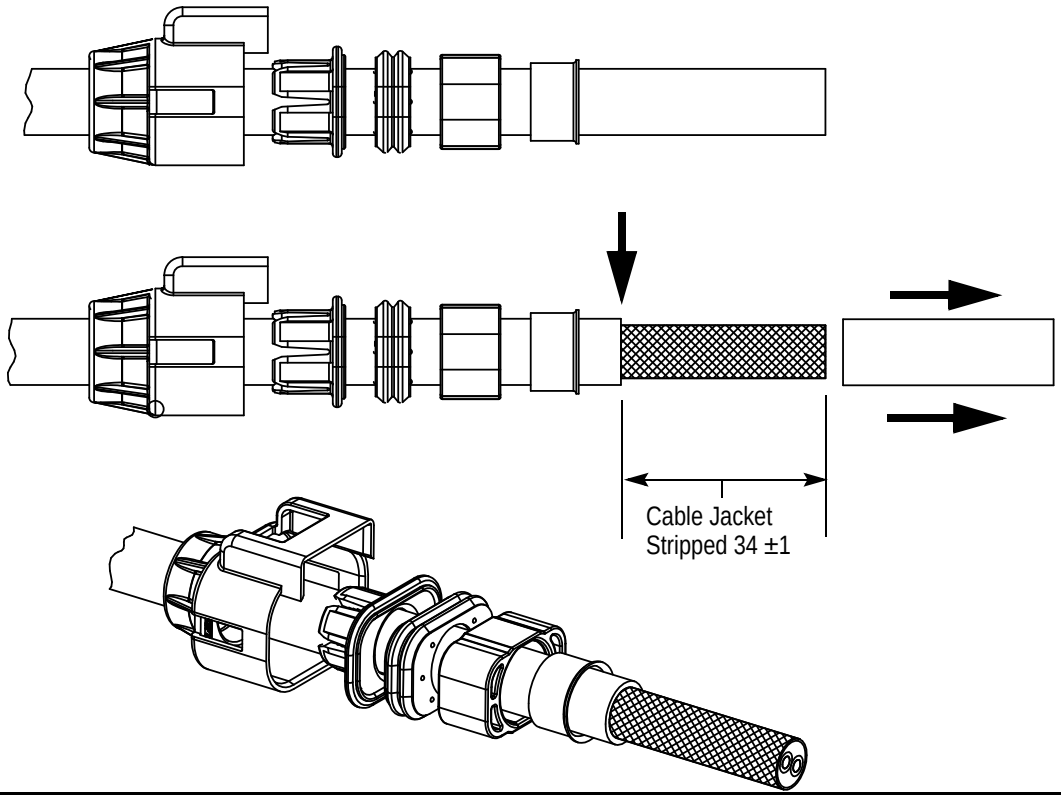


Figure 4

3. Insert the inner ferrule over the cable shield until it stops against the outer jacket as shown in Figure 5.

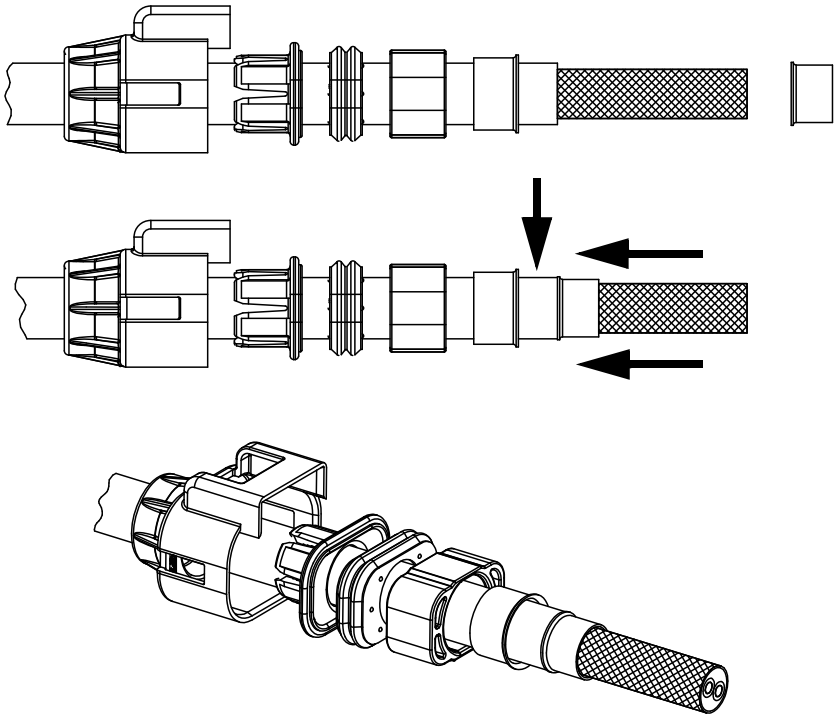


Figure 5

4. Fold the braided cable shield back over the inner ferrule and trim as shown in Figure 6. Remove filler and separator (if present).

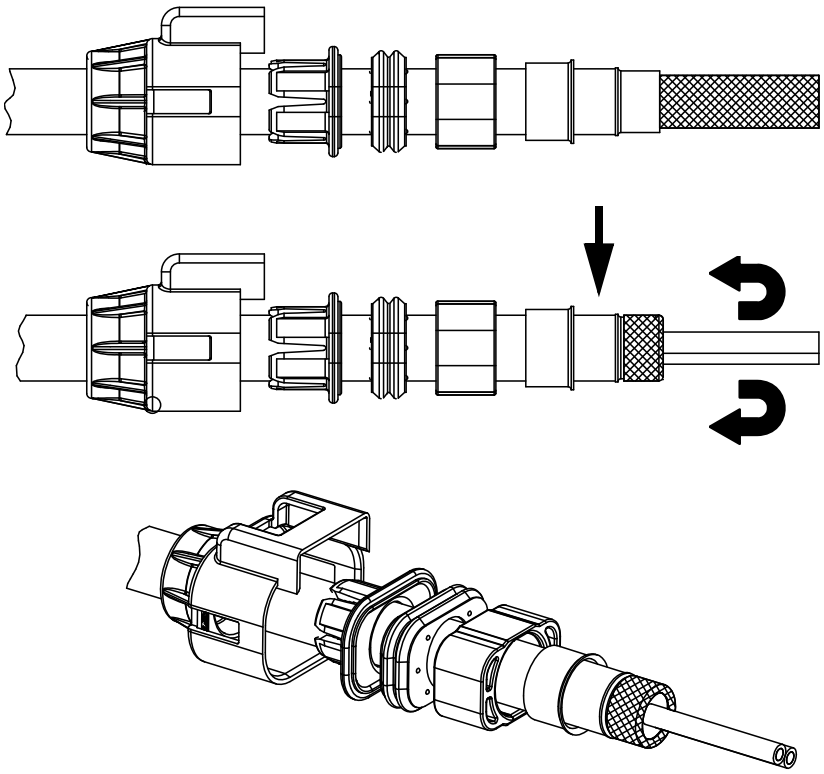


Figure 6

5. Separate conductors and strip to length as shown in Figure 7. Refer to Application Specification 114-18148-1 or 114-18387 for strip length.

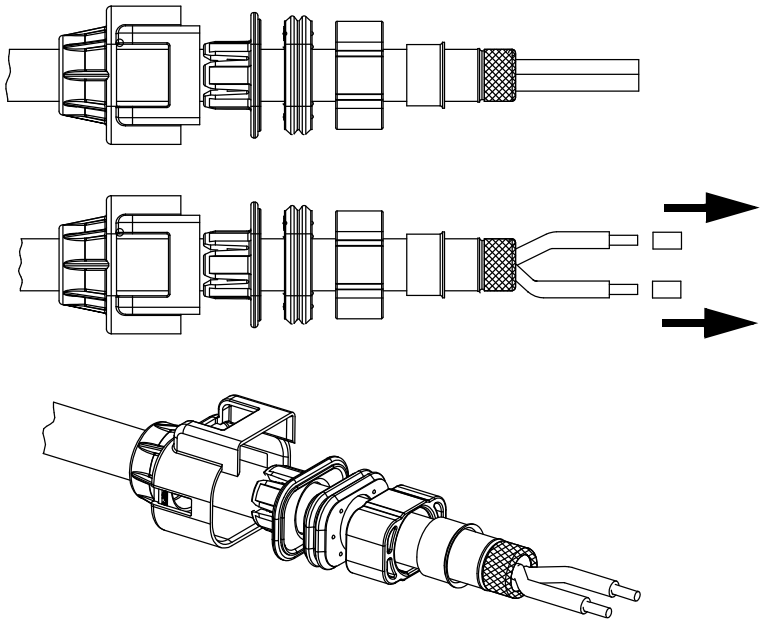


Figure 7

6. Insert the receptacle contacts over the exposed conductor strands and crimp as shown in Figure 8. Refer to Application Specification 114-18148-1 or 114-18387 for specific crimp information. Locate the contact to be crimped in the appropriate tooling according to the instructions packaged with that tooling. Detailed instructions covering the placement of contacts in the tooling and the use of such tooling is packaged with each tool. See Section 5, TOOLING.

CAUTION

Wire insulation shall NOT be cut or broken during the crimping operation, nor shall the insulation be crimped into the contact wire barrel. Reasonable care should be taken by tooling operators to provide undamaged wire terminations.

NOTE

Periodic inspections must be made to ensure crimped contact formation is consistent as shown.

NOTE

Care should be taken to apply the contacts in such a way that minimal strain is required to orient the contact for insertion into the inner housing in the correct order.

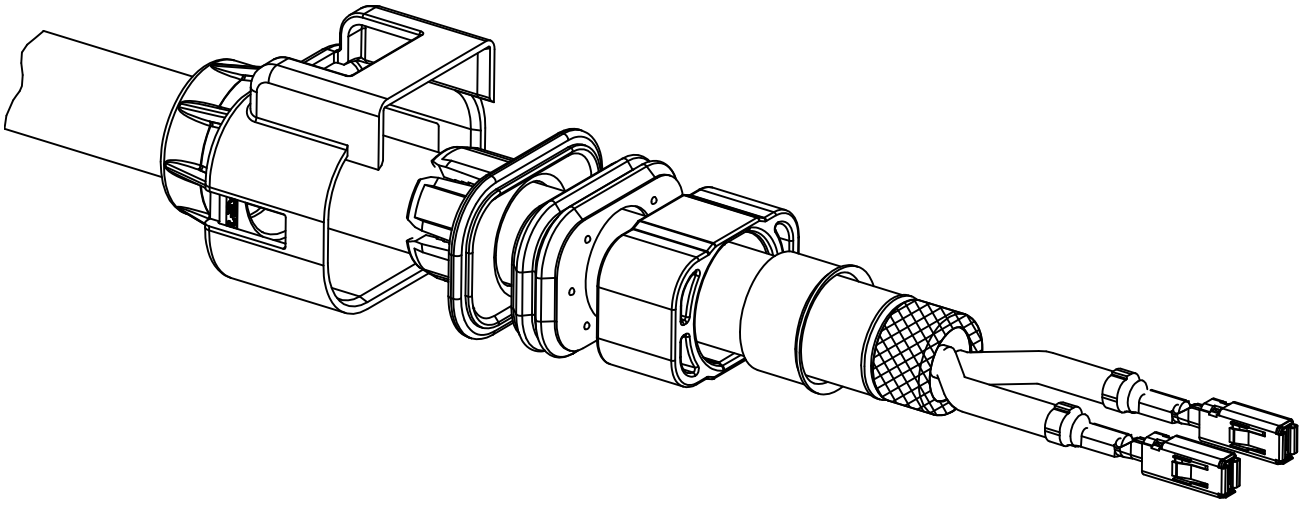


Figure 8

7. The following items at a minimum must be inspected and verified:

- No loose conductor strands
- No loose cable shield strands
- Crimped in correct location
- No excess metallic foil or filler
- All components are present (cable seal retainer, collet, cable seal, spacer, outer ferrule, inner ferrule, receptacle contacts)

8. Pre-bend the conductors as shown in figure 9A to align the terminals with the cavities and prevent damage to the conductor jacket during shield assembly. Note the cavity number on the cable entry end of the inner housing (Figure 9B). Insert the receptacle contacts into the inner housing until they are fully locked. Verify that the contact is completely inserted as shown in Figure 9C by gently pulling back on the conductor.

NOTE

Make sure that the correct conductor color is matched with the correct inner housing cavity number.

CAUTION

It is essential that there are no mistakes in this step because there will be no chance to re-work the parts once the connector shield is crimped in place.

9. *(OPTIONAL)* Two or three wraps of 150° rated tape may be used as shown in Figure 10 to gather the wires if desired. Tape must not overlap onto the braided shield and must remain clear of the inner housing by no less than 8 mm.

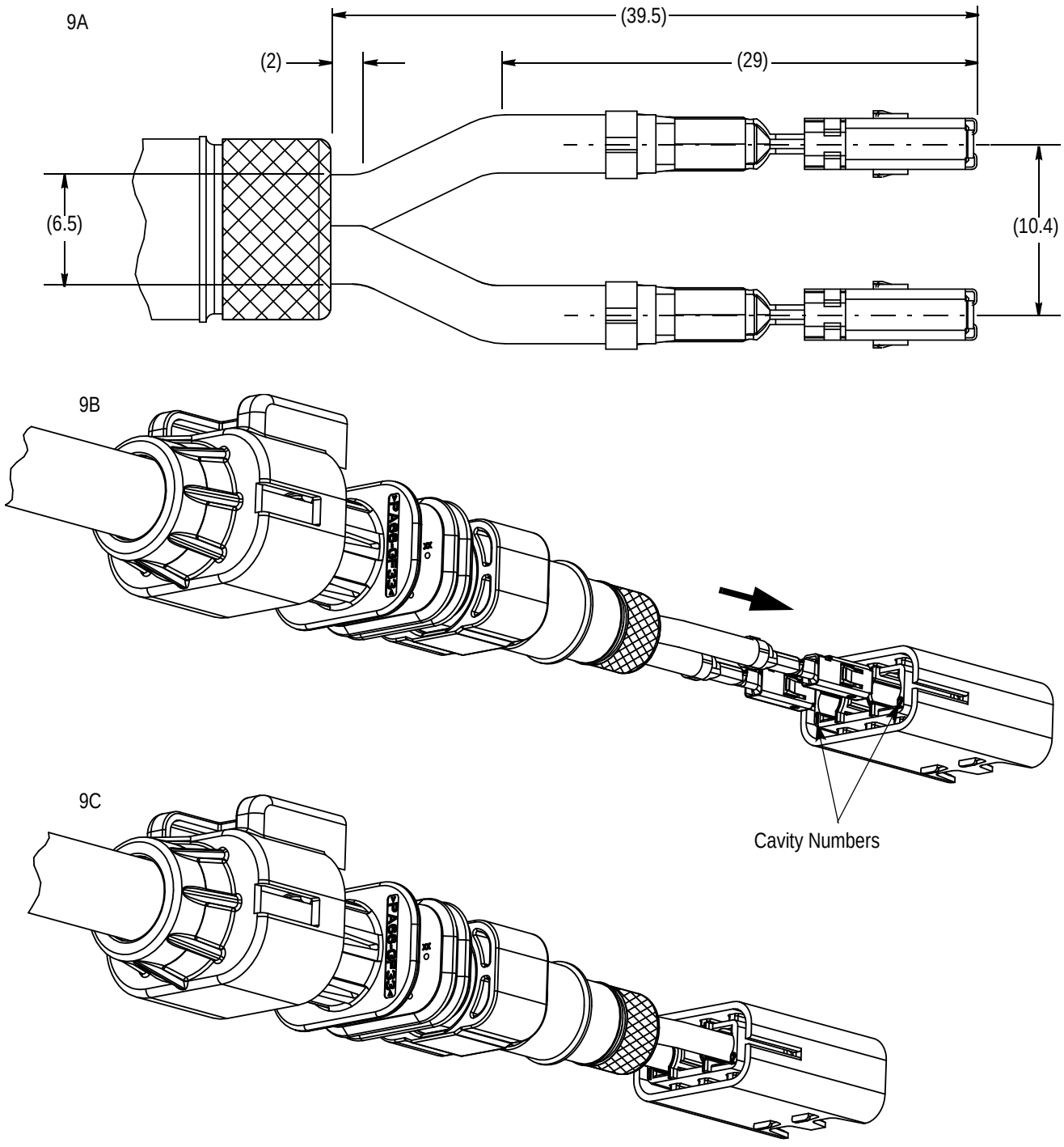


Figure 9

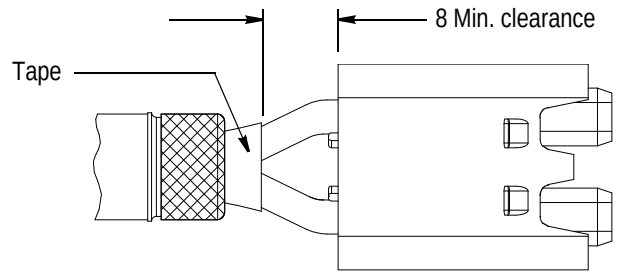


Figure 10

10. Assemble the first connector shield by positioning the inward facing shield tab in the positioning slot of the inner housing on the side shown in Figure 11A. Do not completely lock the other side of this shield at this time. Position the wires away from the shield as shown in the top view of Figure 11A. Assemble the second shield in the forward position of the positioning slot of the inner housing on the side shown in Figure 11B first then rotate until the opposite side shield tab locks into place. Complete the shield assembly process by locking in the other shield tab of the first shield. Make sure the overlapping tabs in the shield transition and crimp area are overlapping correctly and the front shield curl is in the correct position as shown in Figure 11C.

CAUTION

Wire insulation shall NOT be cut or broken during the shield assembly process.

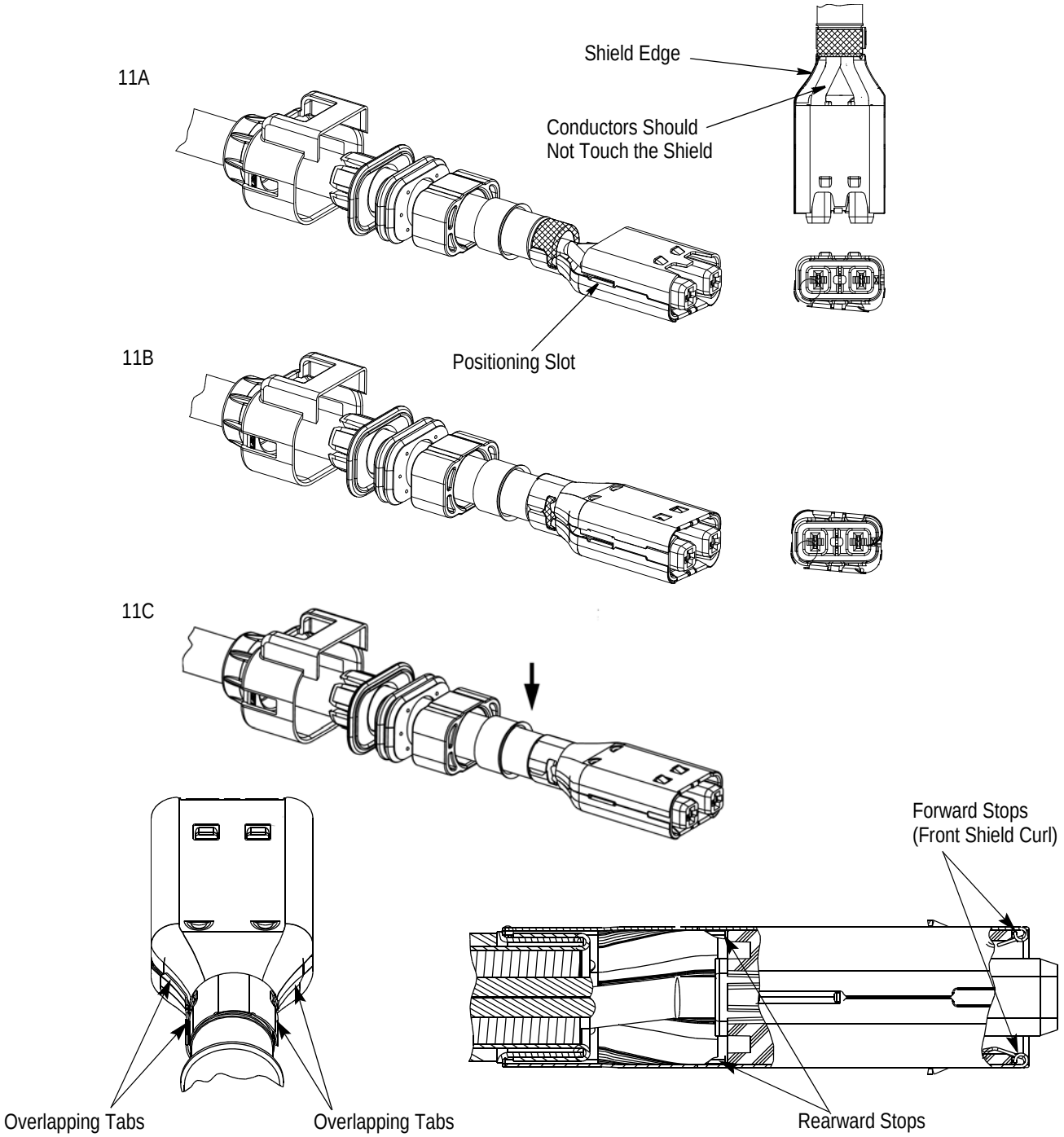


Figure 11

11. Slide the outer ferrule forward into position over the overlapping ends of the connector shields as shown in Figure 12.

12. Crimp outer ferrule to the connector shields in the area shown in Figure 13 using hex crimp tooling as shown in Figure 24.

13. The following items at a minimum must be inspected and verified:

- Hex crimp dimensions per Figure 14
- No large burrs/flash on the ferrule crimp
- No visible cracking of the ferrule or shields
- No loose or detached braid strands
- The cable insulation in close proximity to the ferrule crimp is not pinched, abraded, or cut
- The cable insulation is up against the inner ferrule

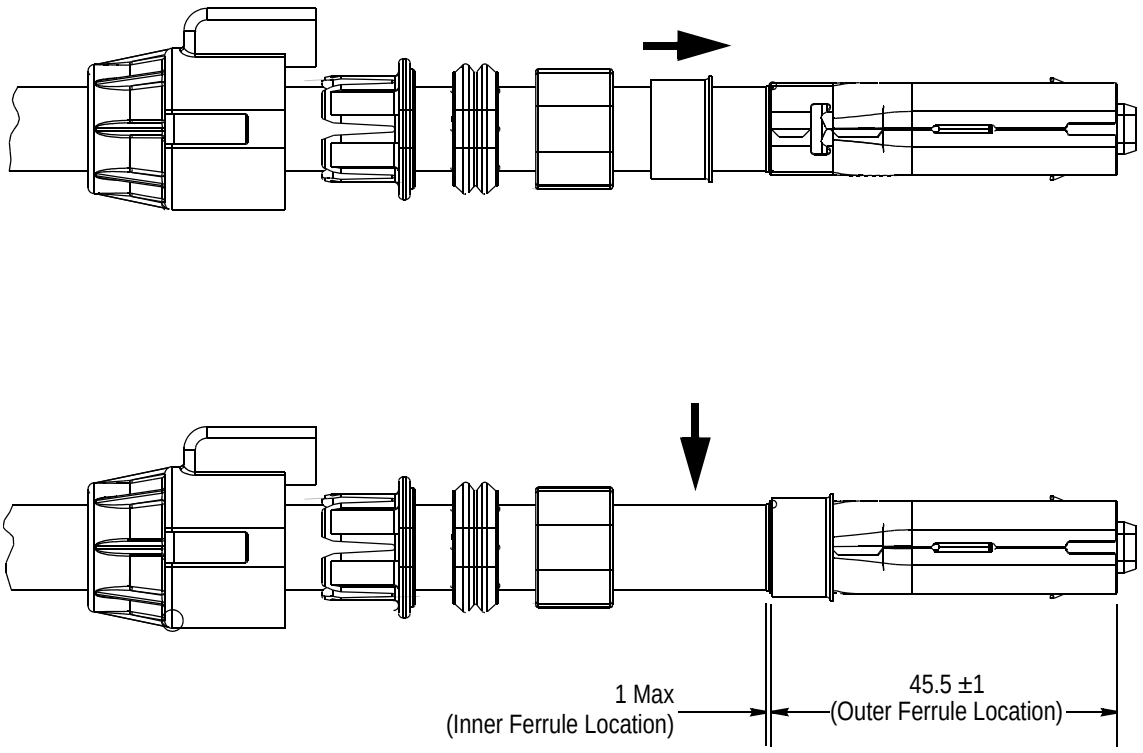


Figure 12

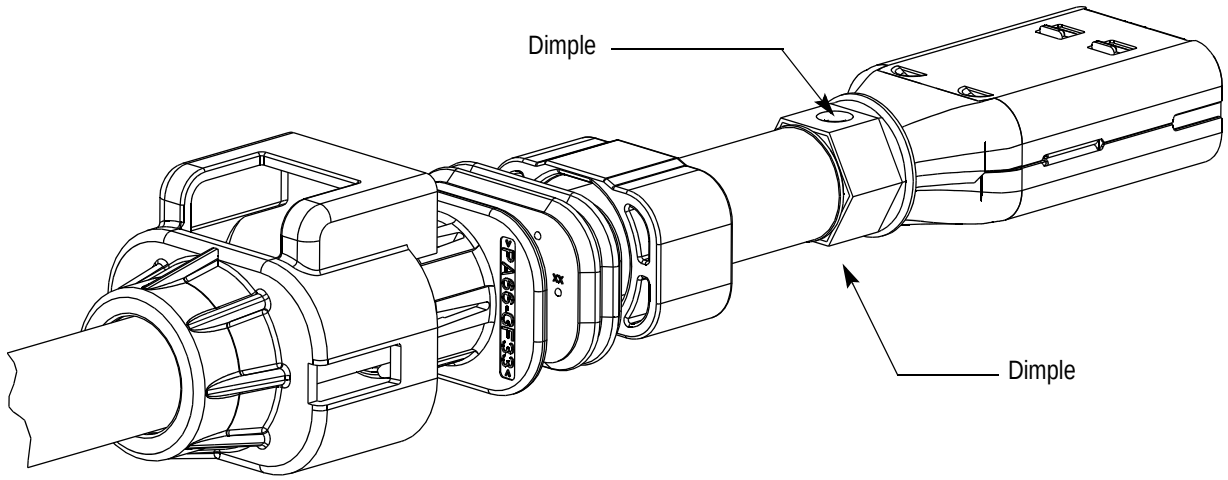
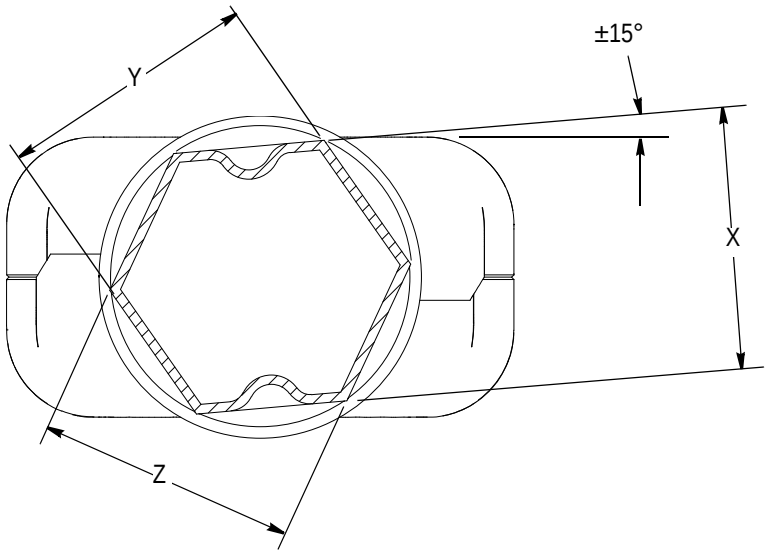


Figure 13

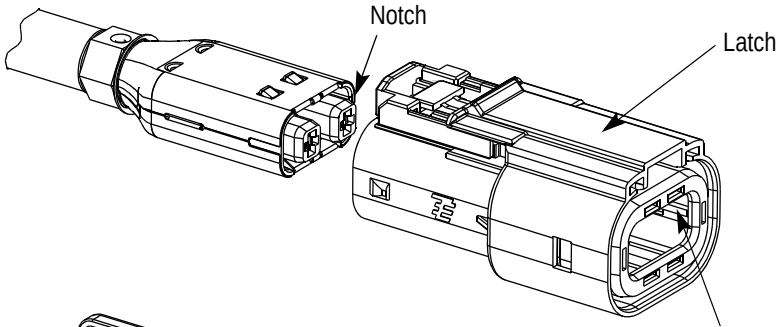


CABLE	DIM X ±0.1	DIM Y	DIM Z
COROPLAST (2X 4.0 mm^2) 9-2641	11.8	11.95 Max	11.95 Max
COROPLAST (2X 2.5 mm^2) 9-2641	10.8	10.95 Max	10.95 Max
DRAKA (2X 4.0 mm^2) FLR91XBC33X			

Figure 14

14. Note the notch on the inner housing subassembly and the rail on the plug assembly housing (Figure 15A). With the notch and rail aligned, insert the inner housing assembly into the plug subassembly until it is fully locked. Verify that the inner housing assembly is completely inserted as shown in Figure 15B by gently pulling back on the cable.

15A



15B

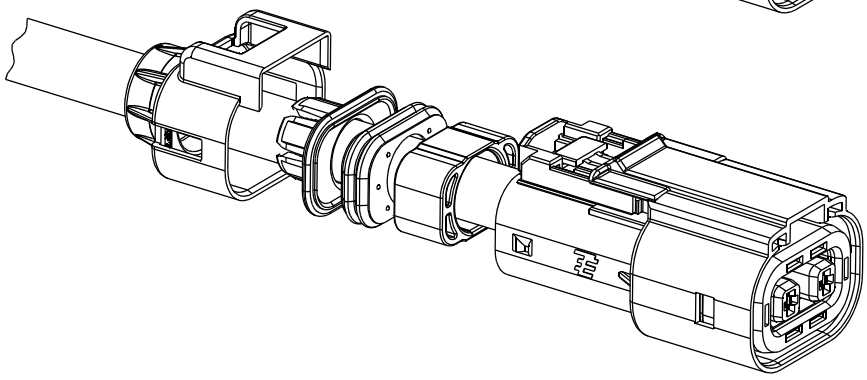


Figure 15

15. Slide the spacer forward on the cable and into the cable exit end of the connector assembly until it stops against the inside of the connector outer housing as shown in Figure 16.

COROPLAST and DRAKA are trademarks of their respective owners.

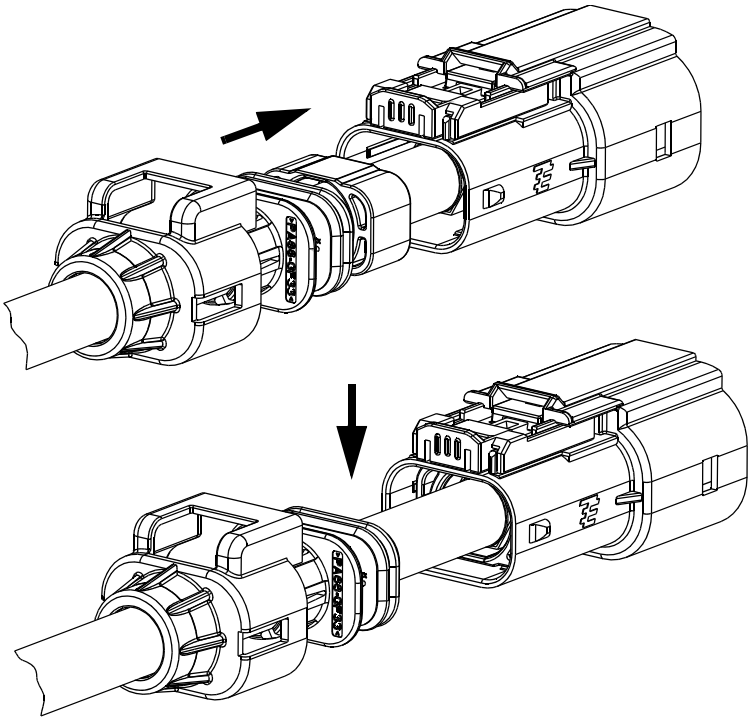


Figure 16

16. Slide the cable seal forward on the cable and into the cable exit end of the connector assembly until it stops against the spacer as shown in Figure 17.

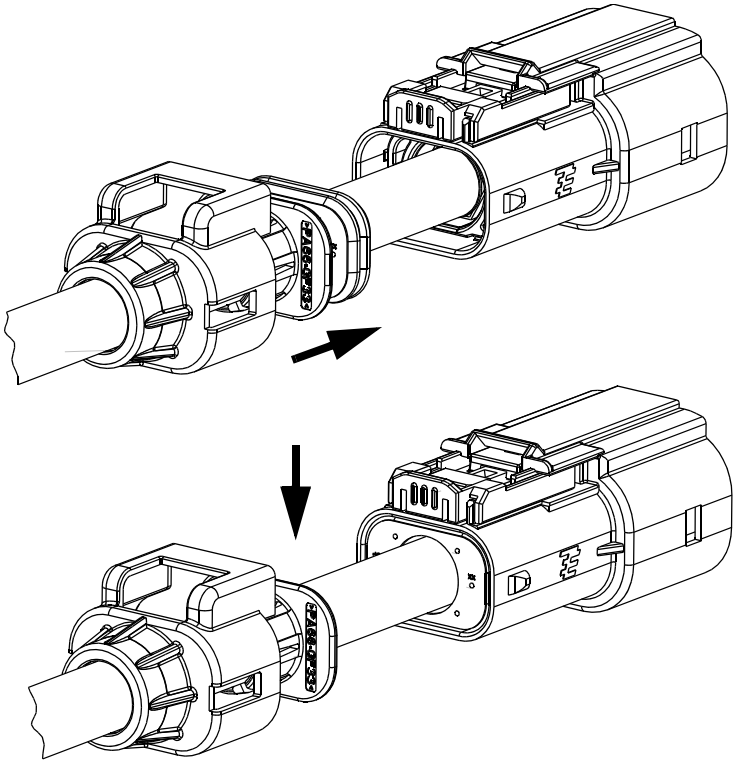


Figure 17

17. Slide the collet forward on the cable until it stops against the outer housing as shown in Figure 18.

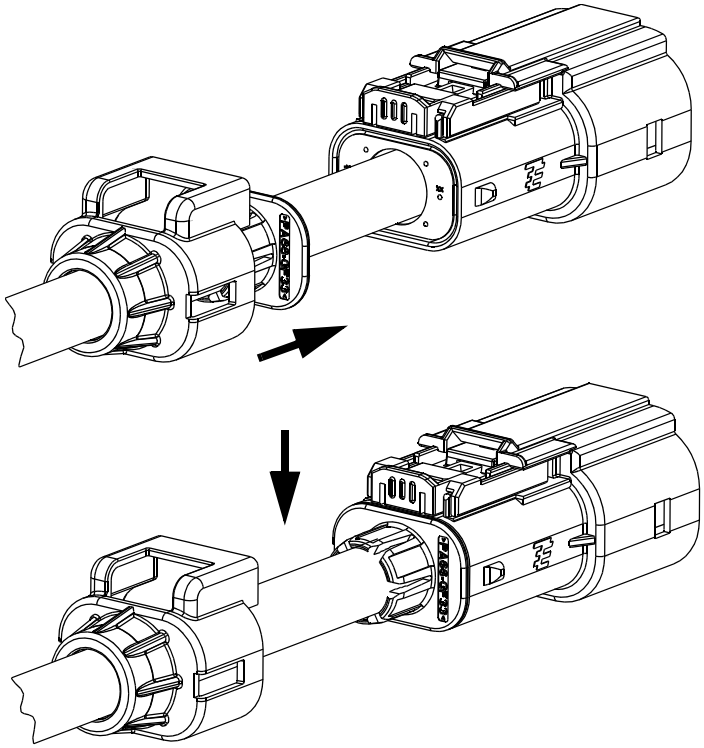


Figure 18

18. Slide the cable seal retainer forward on the cable and over the cable exit end of the connector assembly until it fully locks to the connector outer housing. A slight rotation of the cable seal retainer may be required to get the bridge portion over the CPA. Visually check to ensure that both locking features are engaged and the CPA is in the correct position as shown in Figure 19. Gently pull back on the cable seal retainer to check that it is fully locked in place.

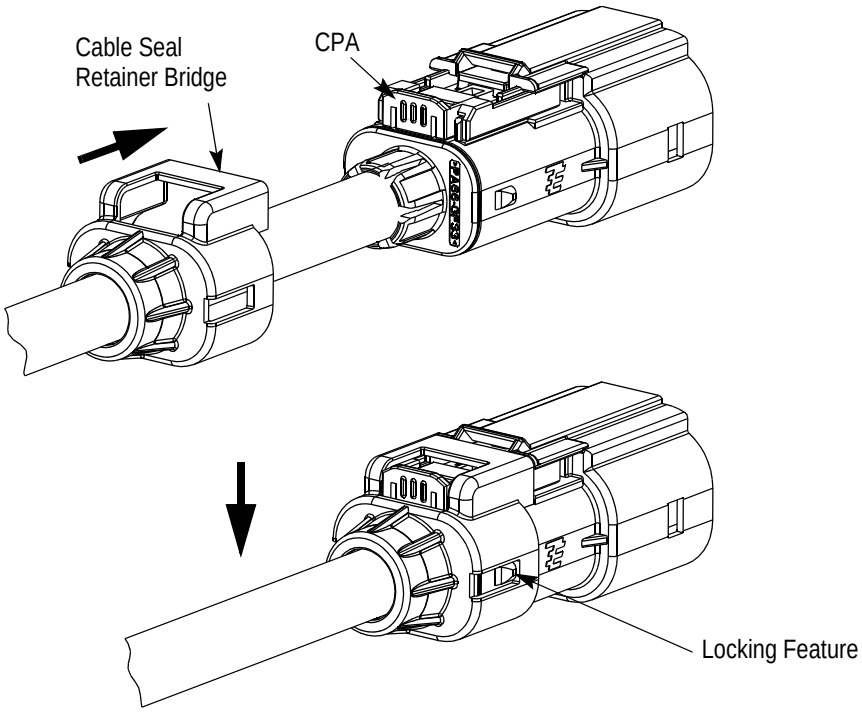


Figure 19

3.7. Contact Probing

CAUTION

Avoid probing the contact in an area that could damage the plating inside the receptacle contact. The best location for probing the receptacle contact is at the front as shown in Figure 20.

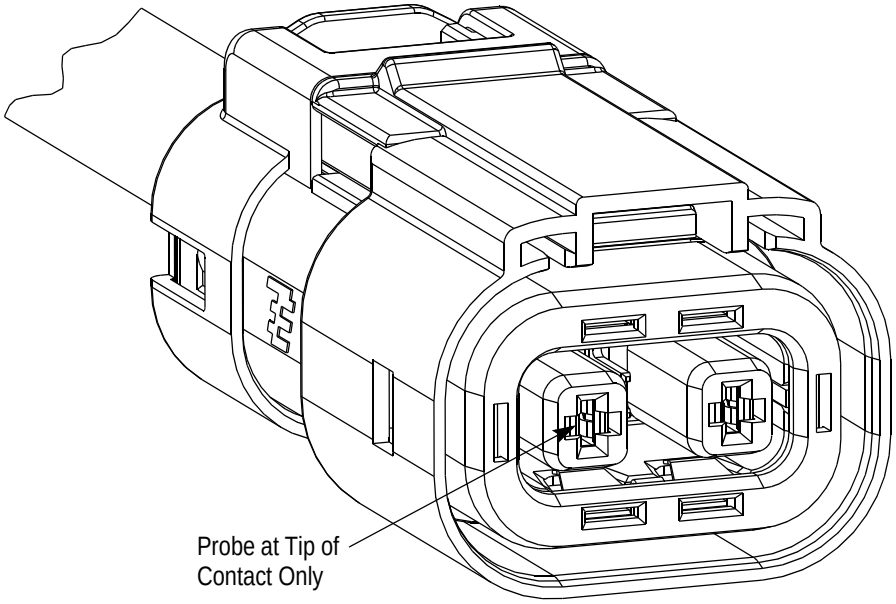


Figure 20

3.8. Connector Mating

1. Align plug and header connectors as shown in Figure 21A.

2. Push the connector halves together fully until they stop (two clicks will be heard).

3. Verify that the connectors are fully locked with a light tug and by visually verifying the thumb actuation pad has returned to its' upward position as shown in Figure 21B.

4. If present, push the red Connector Position Assurance (CPA) forward from the position shown in Figure 21B until it stops as shown in Figure 21C.

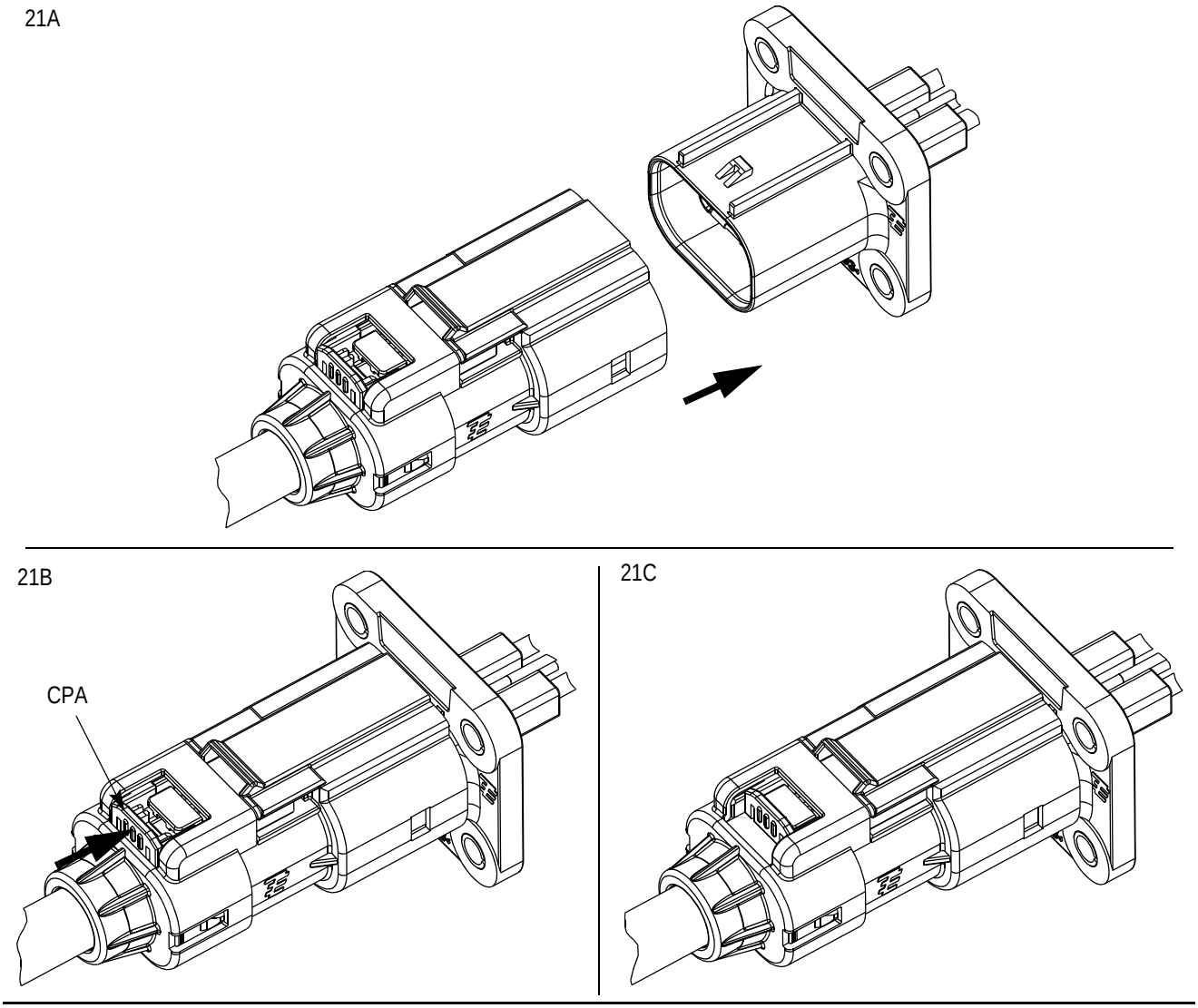


Figure 21

3.9. Connector Unmating

1. If present, pull the red CPA back from the position as shown in Figure 22A until it stops as shown in Figure 22B.

2. Fully depress the thumb actuation pad on the plug connector flexible latch (orange) shown in Figure 22B and pull the plug connector back (~6.2 mm) into the intermediate position. See Figure 22C. At this point, the HVIL circuit has been opened, but the HV terminals are still making contact.

CAUTION

Do NOT pull on the wires.

3. Fully depress the actuation pad on the floating latch (black). See Figure 22D.

4. An alternate construction may not require a flat-bladed screwdriver. See Figure 22E.

5. Pull the plug connector until it is fully separated from the header. See Figure 22F.

CAUTION

Do NOT pull on the wires.

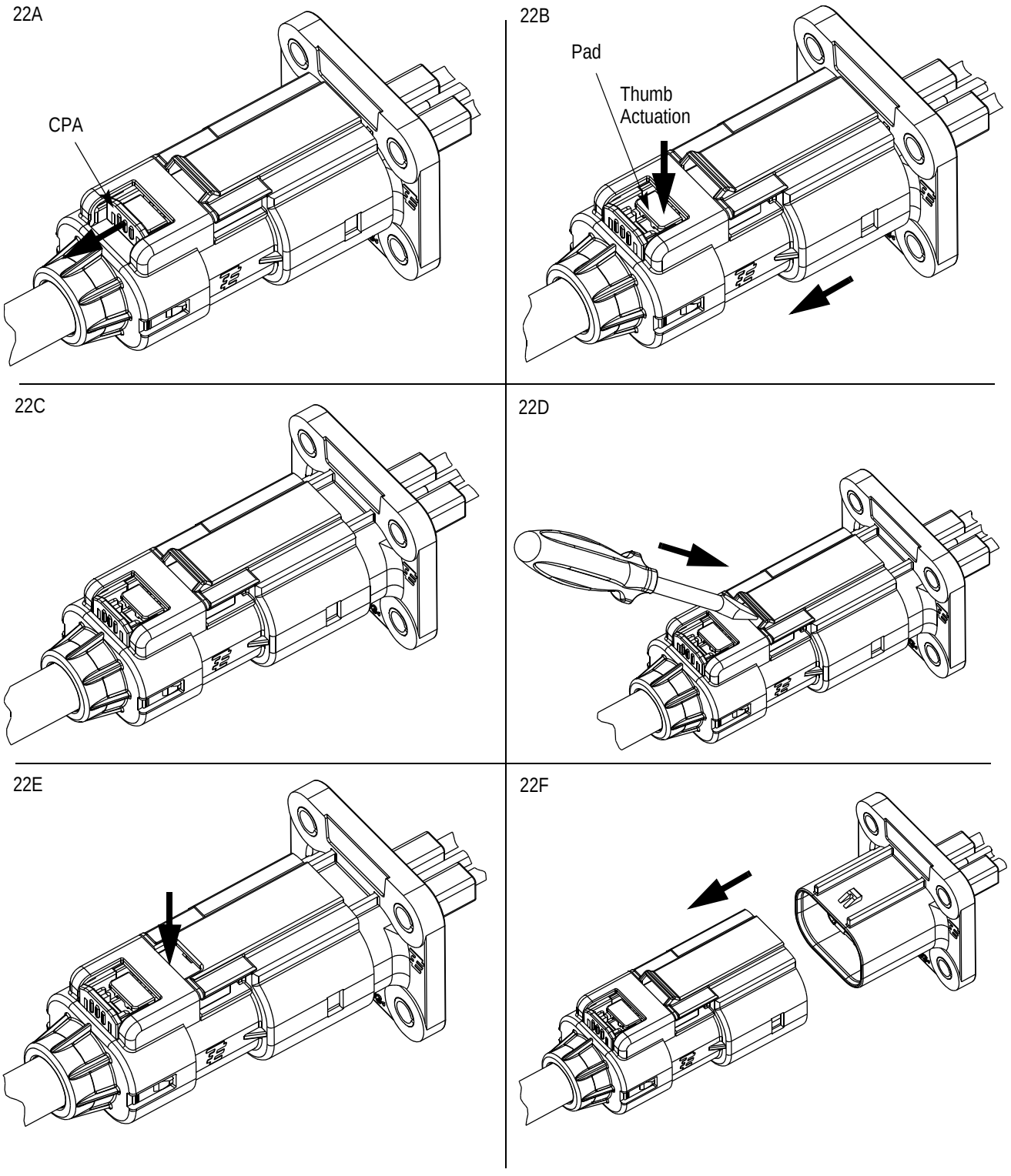


Figure 22

3.10. Cable Routing

Cable assembly should be restrained as shown in Figure 23.

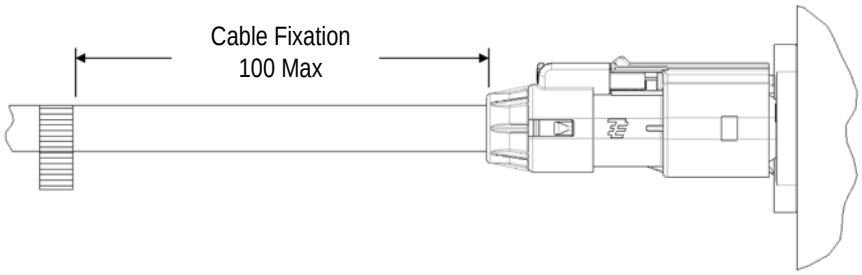


Figure 23

4. QUALIFICATIONS

The AMP+ HVA280 connector system has been validated to the following specifications:

- IEC 60528: Degrees of Protection Provided by Enclosures (IP Code)

5. TOOLING

A listing of tooling recommendations covering the full wire size range is provided in Figure 24. Modified designs and additional tooling concepts may be available to meet other application requirements. For additional information, contact one of the service groups at the bottom of page 1.

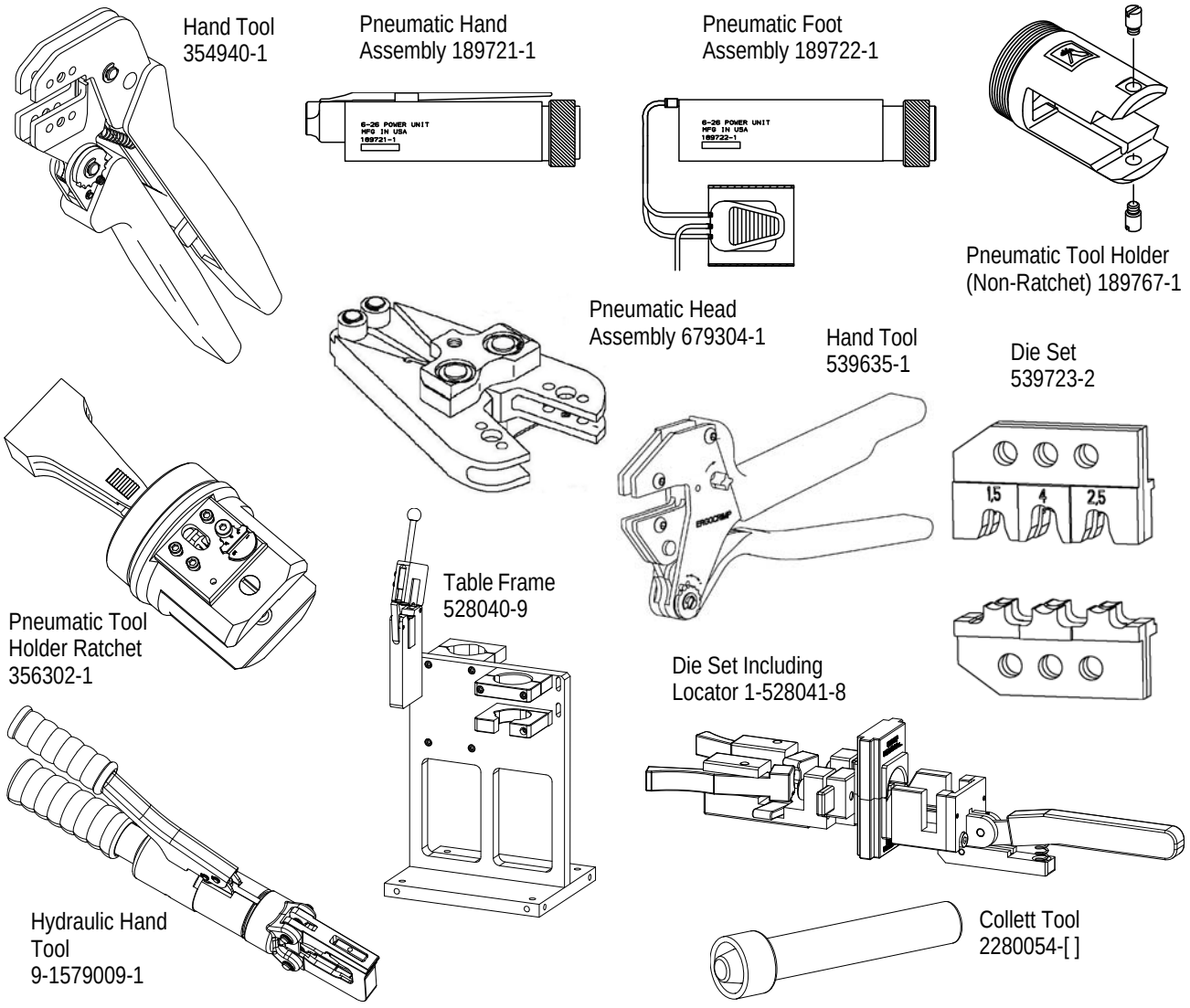


Figure 24 (Cont'd)

APPLICATION TOOLING FOR FERRULES (DOCUMENT)										
CABLE	DIE SET	MANUAL TOOLING (408-9930)	PNEUMATIC TOOLING					HYDRAULIC TOOLING		
			HAND ASSY (408-2498) (409-5862)	FOOT ASSY (408-2498) (409-5862)	TOOL HOLDER			DIE SET INCLUDING LOCATOR	HAND ASSY PLIERS	TABLE FRAME FOR PLIERS
					RATCHET (408-4303)	NON-RATCHET (408-2498)	HEAD ASSY (408-4070)			
COROPLAST (2X 4.0 mm²) 9-2641	2063013-7	Hand Tool Frame Assy 354940-1†	189721-1	189722-1	356302-1	189767-1	679304-1	1-528041-8	9-1579009-1	528040-9
COROPLAST (2X 2.5 mm²) 9-2641	2063013-5	Hand Tool Frame Assy 354940-1						2-528041-3		
DRAKA (2X 4.0 mm²) FLR91XBC33X										

APPLICATION TOOLING FOR 1-968851-3 and 1-968853-3 RECEPTACLE CONTACTS			
CONTACT TYPE	WIRE SIZE RANGE	HAND TOOL (411-18087)	DIE SET (411-18169)
Receptacle	2.5 mm²	539635-1	539723-2
	4.0 mm²		

APPLICATION TOOLING FOR 1241390-3 RECEPTACLE CONTACTS			
CONTACT TYPE	WIRE SIZE RANGE	HAND TOOL (411-18087)	DIE SET (411-18138)
Receptacle	2.5 mm²	539635-1	539951-2

APPLICATION TOOLING FOR COLLET ASSEMBLY	
COLLET PART NUMBER	COLLET TOOL PART NUMBER (408-32128)
2103155-1	2280054-1
2103155-3	2280054-3

†With modification.

Figure 24 (End)

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6. VISUAL AID

The illustration below shows a typical application of this product. This illustration should be used by production personnel to ensure a correctly applied product. Applications which DO NOT appear correct should be inspected using the information in the preceding pages of this specification and in the instructional material shipped with the product or tooling.

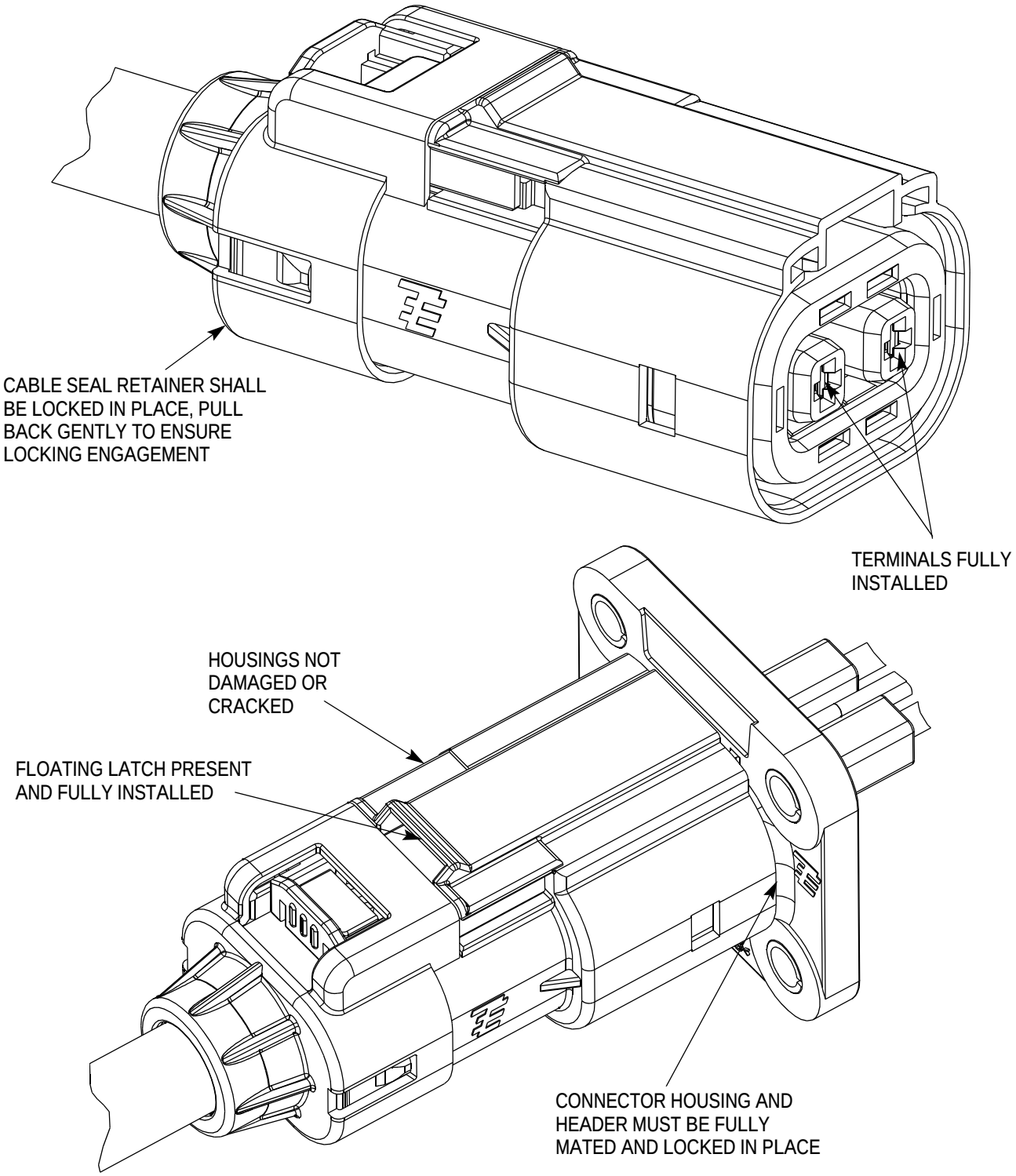


FIGURE 25. VISUAL AID